

Following the mathematical definition of covariance and correlation we find that the covariance of price change between two stocks is really a measure of the co-movement of the “error terms” of each stock not the co-movement of prices. For example, the statistical definition of covariance between two random variables x and y is:

$$\sigma_{xy} = E[(x - \bar{x})(y - \bar{y})]$$

It is quite possible for two stocks to have the same exact trend but whose errors (noise term) are on opposite sides of the trend lines. For example, if $\bar{x} = \bar{y} = z$, $x = d$, $y = -d$ and $d > z$, our covariance calculation is:

$$E[(x - \bar{x})(y - \bar{y})] = E[(d - z)(-d - z)] = E[-d^2 + z^2]$$

Since $d > z$, we have $E[-d^2 + z^2] < 0$ which is a negative measured covariance term indicating the stocks trend in opposite directions. But these two stocks move in exactly the same direction, namely, z .

It is also possible for two stocks to move in opposite directions but have a positive covariance measure. For example, if $\bar{x} = z$ and $\bar{y} = -z$, $x = y = d$, and $d > z$, the covariance calculation is:

$$E[(x - \bar{x})(y - \bar{y})] = E[(d - z)(d - -z)] = E[d^2 - z^2]$$

Since $d > z$ we have $E[d^2 - z^2] > 0$ which is a positive measured covariance term indicating the stocks trend in the same direction. But these two stocks move in the exact opposite direction.

The most important finding above is that when we compute covariance and correlation on a stock by stock basis using historical returns and price data it is possible that the calculated measure is opposite of what is happening in the market. These “false positive” and/or “false negative” relationships may be due to the error term about the trend rather than the trend or possibly due to too few data points in our sample.

Example 1: False negative signal calculations

Table 6.5a contains the data for two stocks A and B that are moving in the same direction. Figure 6.7a illustrates this movement over 24 periods. But when we calculate the covariance between these stocks we get a negative correlation, $\rho = -0.71$. How can stocks that move in the same direction have a negative covariance term? The answer is due to the excess terms being on opposite sides of the price trend (Figure 6.7b). Notice that these excess returns are now on opposite sides of the trend which results in a negative covariance measure. The excess returns are indeed negatively correlated but the direction of trend is positively correlated.